

VISANATOMY: An SVG Chart Corpus with Fine-Grained Semantic Labels

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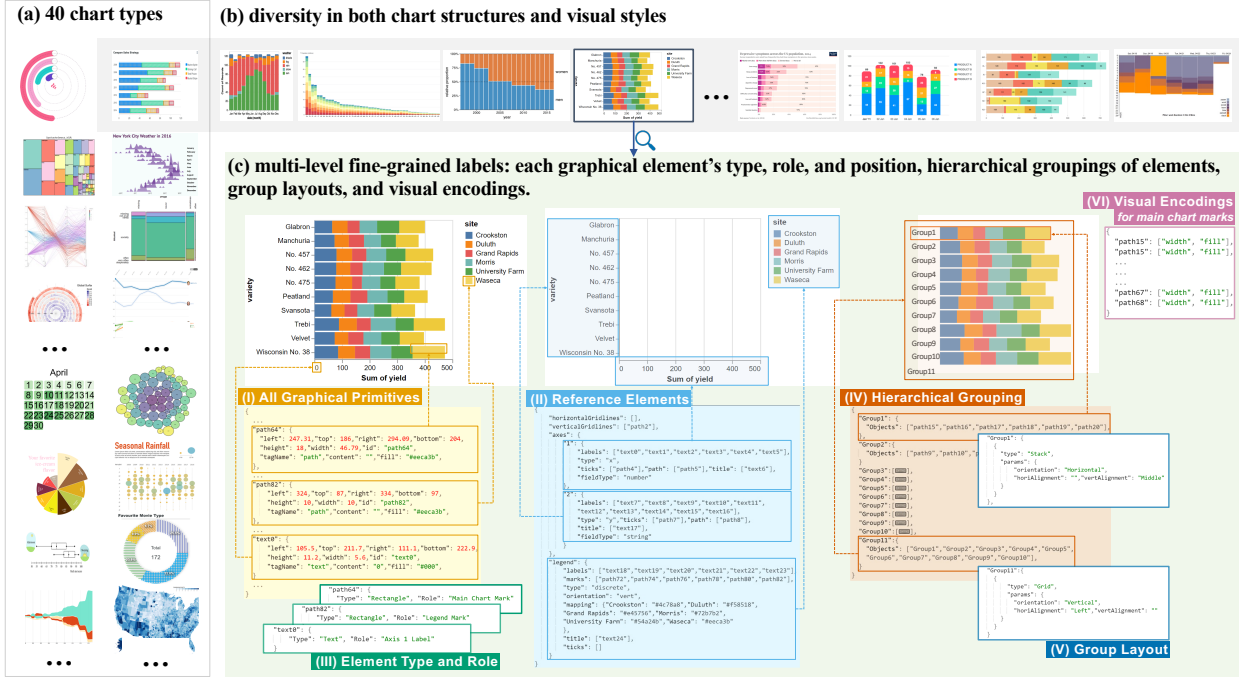


Fig. 1: VISANATOMY is an SVG chart corpus that (a) consists of 942 charts across 40 chart types, (b) promotes diversity within each chart type, featuring rich design variations in terms of visual structures and styles, and (c) distinguishes from existing chart corpora in its multi-granular chart semantic labels on more than 383k graphical elements, including each element's type, role, and position, hierarchical groupings of elements, group layouts, and visual encodings.

Abstract—Chart corpora, which comprise data visualizations and their semantic labels, are crucial for advancing visualization research. However, the labels in most existing corpora are high-level (e.g., chart types), hindering their utility for broader applications in the era of AI. In this paper, we contribute VISANATOMY, a corpus containing 942 real-world SVG charts produced by over 50 tools, encompassing 40 chart types and featuring structural and stylistic design variations. Each chart is augmented with multi-level fine-grained labels on its semantic components, including each graphical element's type, role, and position, hierarchical groupings of elements, group layouts, and visual encodings. In total, VISANATOMY provides labels for more than 383k graphical elements. We demonstrate the richness of the semantic labels by comparing VISANATOMY with existing corpora. We illustrate its usefulness through four applications: semantic role inference for SVG elements, chart semantic decomposition, chart type classification, and content navigation for accessibility. Finally, we discuss research opportunities to further improve VISANATOMY.

Index Terms—Chart, SVG, data visualization, corpus, dataset, multilevel fine-grained semantic labels

1 INTRODUCTION

Visualization researchers have been curating chart corpora to advance the state of the art in chart creation and generation [28, 22], classifi-

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cation [39, 66], retrieval [46, 36], decomposition [15, 61], and editing [15, 21]. The availability of fine-grained semantic labels such as element properties and data encodings is vital for a range of visualization downstream tasks. For example, the shapes and roles of visual elements as well as their visual properties are required to develop (semi-)automated data visualization reuse pipelines [15, 66]; the correspondence between visual elements (or groups) and axis/legend labels is necessary for developing chart reader experiences for visually impaired people [79]; the grouping structure of visual elements can be utilized to develop graph neural network models [46].

However, the semantic labels in existing chart corpora are often insufficient and sometimes unreliable for supporting various visualization tasks [15, 16]. According to two recent surveys [16, 25], many corpora are not publicly available. The remaining ones typically have

only high-level labels (e.g., chart types [8, 66] and chart area bounding boxes [27]). Although a few existing corpora do offer fine-grained labels, they often exhibit limited diversity in terms of the variety of chart-authoring tools and chart designs. This limited diversity hinders the generalizability of models built upon those corpora: they can easily fail when handling “out-of-distribution (OOD)” charts produced by other tools with inconsistent usage of SVG elements [46] and grouping structures [15]. In general, existing chart corpora are inadequate to support the development of robust visualization applications.

In this paper, we seek to address these limitations and contribute a diverse SVG chart corpus, VISANATOMY, with multi-level fine-grained semantic labels. VISANATOMY includes 942 real-world SVG charts and their corresponding multi-level semantic labels. The underlying data tables are also included if available (329 out of 942). The charts are collected through a manual process with careful inspection. For each chart, multiple independent expert annotators use a semi-automated annotation tool to obtain the semantic labels, and the quality of the labels is controlled through consensus among them.

VISANATOMY makes two key extensions over prior chart corpora. First, regarding corpus diversity, VISANATOMY encompasses 40 chart types (synthesized from three visualization typologies [14, 63, 30]) produced by over 50 tools from hundreds of public online sources, featuring structural and stylistic design variations (Section 2). Second, and more importantly, we identified a set of core components through a survey on existing visualization scene models [48, 64, 65, 69] to augment each chart in VISANATOMY with comprehensive semantic labels, including each visual element’s shape (e.g., rectangle, pie, polyline), role (e.g., main chart mark, axis path, legend tick, annotation), and bounding box, the hierarchical grouping of elements, the layout for each group, and visual encodings (Section 3.1). In total, VISANATOMY provides labels for 383,459 visual elements.

We evaluate the richness of semantic labels and corpus diversity through a comparison between VISANATOMY and existing corpora (Section 3.4). Four applications illustrate the usefulness of VISANATOMY (Section 4): semantic role inference for SVG elements, chart semantic decomposition, chart type classification, and content navigation for accessibility. Finally, we discuss the current limitations of VISANATOMY and outline our future work (Section 5). The VISANATOMY corpus is available at <https://VisAnatomy.github.io/>.

2 VISANATOMY: CHART COLLECTION

In this section, we introduce the process to construct VISANATOMY. We first describe how the team decided on the desired chart types and collected charts following a standardized procedure. We then give an overview of the 942 charts in VISANATOMY, showing the distributions of the chart types, charting tools, and source domains.

2.1 Manual Chart Collection

Before the collection process started, we decided to focus on charts in the SVG (Scalable Vector Graphics) format. In recent years, SVG has emerged as a popular choice for curating corpora with fine-grained semantic labels in diverse visualization applications [15, 46, 53, 61, 69]. Compared to raster images, SVG includes low-level details such as element types and visual styles in its XML structure [16], removing the need for error-prone image segmentation [57] and element extraction [56]. Compared to code, where a label extraction approach is not easily generalizable to different visualization languages [55], SVG is supported as the output format by a wide array of languages and tools.

We manually searched for and sampled real-world SVG charts online to form the chart collection in VISANATOMY. There are other approaches to collecting charts, such as web crawling, that could lead to larger corpora. However, charts collected through automatic crawling cannot guarantee a balanced distribution of charts or diversity in terms of chart types and design variations [16]. Also, since SVGs can be embedded in web pages in various ways (e.g., inline SVG, object tag, iframes), consistent extraction of SVG charts in practice is more challenging compared to raster images, which are mostly directly embedded. We therefore decided to follow a manual collection process to ensure the quality and diversity of the SVG charts.

The first step in our chart collection process was to compile a set of targeted chart types by browsing three visualization typologies: the Chartmaker Directory [14], the Data Viz Project [30], and the Data Visualisation Catalogue [63], each of which contains a detailed categorization of chart types. We cross-checked the named chart types in these three collections and focused on visualizations that consist of basic geometric shapes including rectangle, circle, ellipse, pie, arc, line, polyline, area, polygon, geo-polygon, and text. For the underlying data in the visualizations, Munzner [58] defined four types of datasets: tables (items & attributes), trees and networks (nodes & links), fields, and geometry. We decided to focus on visualizations of tables, thus excluding node-link diagrams and scientific visualizations. Applying these criteria, we narrowed down to 31, 37, and 37 chart types from the three typologies, respectively. Finally, we unified the names of these chart types and achieved a final set of 40 chart types (Figure 2).

We then started collecting SVG charts for each chart type by (1) browsing online charting tool galleries (e.g., D3.js [11], Vega-Lite [65], Mascot.js [49]), (2) browsing online communities where visualizations are shared by chart makers (e.g., Observable [60], bl.ocks.org [10], Spotfire [70]), and (3) searching for certain chart types using engines such as Google Advanced Image Search [1] (with “SVG” specified as the target file type) and Bing Visual Search [3]. It is important to note that when collecting charts, we *prioritized diversity over quantity because promoting rich design variations and supporting broader visualization applications are our research goals*. Therefore, for each chart type, we focused on including designs from different visualization galleries and websites, showcasing diverse visual styles. We aimed to include at least 20 designs for each chart type. However, we kept adding new designs as long as they introduced unique visual styles that had not yet been observed and did not distort the overall distribution of chart types (i.e., an approximately uniform distribution). We also examined the details of SVG files to discard invalid ones, e.g., those using <image> elements with hyperlinks to render the whole chart. When the appearance of an SVG is influenced by webpage style rules (i.e., CSS), we would first attempt to manually inject the relevant styles into the SVG. If that was not feasible, the SVG was discarded. Note that we mainly focused on styles vital for chart integrity (e.g., color, stroke), and did not handle all the font-related styles in this process. For each SVG chart collected, we obtained its image in the PNG format with a Python script.

The project team consisting of six authors held weekly meetings to inspect the collected SVG charts, removed unqualified charts (e.g., repeated or highly similar charts), and enforced consistent selection criteria. Charts that do not fall into one of the 40 types but still embody interesting design ideas were moved to the “Others” category where we store bespoke chart designs. This process was repeated until we had the expected number of valid designs (i.e., 20) for each chart type. This iterative process ensures that the whole team has at least one pass on every collected chart and that issues with collected charts are resolved consistently. After a 28-week chart collection effort, we reached the final set of charts in both the vector and raster image formats in VISANATOMY, containing a total number of 942 charts with an approximately even distribution across the 40 chart types plus an “Others” category, encompassing more than 50 charting tools and hundreds of web domains (chart sources).

2.2 VISANATOMY Promotes Chart Diversity

Chart Types. In its current state, VISANATOMY contains 942 real-world SVG charts. Each chart has its corresponding label file in the JSON format, and is also available in the PNG (Portable Network Graphics) format. Figure 2 shows the overall distribution of the chart types categorized by the mark types. Within VISANATOMY, there are 40 named chart types together with an “Others” category containing custom designs such as composite or superimposed charts that do not fall appropriately into a specific chart type. Line graph, area chart, bar chart, grouped bar chart, and the “Others” category constitute higher proportions in the corpus, and the number of charts in each remaining type ranges between 20 to 26. This distribution makes VISANATOMY

a balanced corpus with a wide variety of chart types.

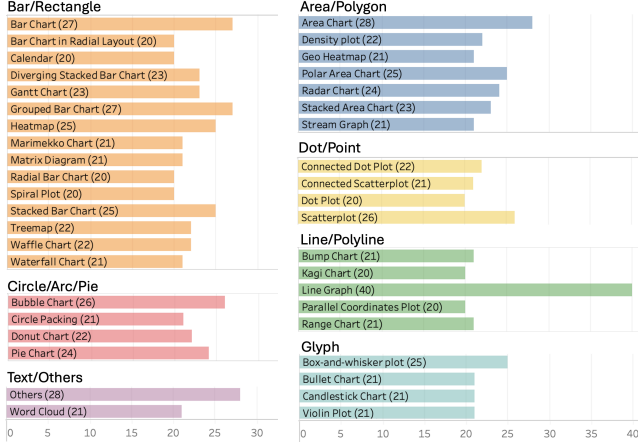


Fig. 2: Chart type distribution in VISANATOMY categorized by primary mark types. Within each category the chart types are sorted alphabetically.

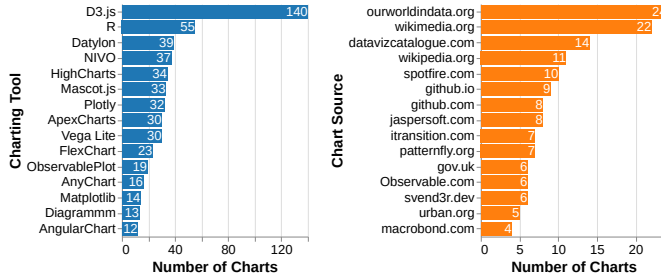


Fig. 3: Distributions of chart types, charting tools (top-15), and source domains (top-15) in VISANATOMY.

Charting Tools and Chart Sources. For each chart, we record information on the charting tool (if it is explicitly revealed in the source website) and/or the website domain (if it is not coming from a charting tool’s gallery). In total, VISANATOMY collects charts created using 58 charting tools and more than 100 different online domains. In Figure 3 we show the top 15 charting tools and chart sources. D3.js [11] contributes the largest portion as it is the most expressive tool for creating interactive web-based SVG visualizations [8]. Several other visualization grammars and tools also provide a decent amount of charts to VISANATOMY, including R [75], Datylon [24], NIVO [59], Highcharts [34], Mascot.js [49], Plotly [2], Apexcharts.js [5], and Vega-Lite [65]. The number of charts is more evenly distributed across chart sources, as most domains occur fewer than five times.

Design Variations. In VISANATOMY, we also strive for chart design diversity within each chart type, featuring rich design variations in terms of both chart structures and visual styles. Figure 1(b) shows nine exemplary design variations for the stacked bar chart type. In VISANATOMY, design variations include but are not limited to:

- different mark types used to create charts of the same type (e.g., bump charts composed of area marks or polylines with dots);
- different orientations of marks (e.g., connected dot plots containing horizontal or vertical glyphs);
- layering of marks (e.g., superimposed area charts);
- nested structures (e.g., small-multiple waffle charts and grouped box and whisker plots);
- different positions, orientations, and visual styles of reference elements such as axes, legends, and gridlines;
- different styles of annotations and embellishments.

We include detailed information of the chart types, tools, and sources in VISANATOMY in the supplementary materials¹.

¹see [detailed information](#) of VISANATOMY’s chart collection.

3 VISANATOMY: MULTILEVEL FINE-GRAINED SEMANTIC LABELS

We took an iterative in-house labeling approach to obtain high-quality semantic labels for a set of core components synthesized from literature [48, 64, 65, 69]: mark elements reference elements, hierarchical grouping, visual encodings, and group layout. We decided not to obtain labels through crowdsourcing to ensure the label quality in VISANATOMY. Crowdsourced labels often require significant time to inspect and correct [43, 41, 78]. Moreover, our desired labels require visualization expertise from the annotators, which is hard to guarantee through crowdsourcing platforms. Three experts in the team, each with at least four years of visualization research experience, participated in labeling the semantic components of the collected SVG charts. In this section, we describe the semantic labels that are associated with the charts in VISANATOMY and the labeling process.

3.1 Fine-grained Labels of Multilevel Scene Components

To support a broad set of visualization applications, we need detailed semantic labels beyond just chart types [16]. Specifically, we need a comprehensive understanding of the structure of a chart at multiple levels of granularity, from its global-level chart type down to the properties of individual elements. To this end, we surveyed related literature to compare existing visualization abstractions [48, 64, 65, 69]. We finally decided to focus on the labels for the following components commonly shared across visualization grammars and abstractions: mark elements, reference elements, hierarchical grouping, visual encodings, and group layout. The supplemental materials contain detailed descriptions of the components we have surveyed.

Using the stacked bar chart presented in Figure 1(c) as an example, we show its scene structure (gray nodes and edges) together with the correspondence to the semantic labels recorded in VISANATOMY in Figure 4. Specifically, **Reference Elements** specify properties and involved elements for gridlines, axes, and legend; **All Graphic Primitives** and **Element Type and Role** contain detailed information about all graphical elements in the scene, including rectangle marks and **Reference Elements**; **Hierarchical Grouping** and **Group Layout** correspond to the spatial clusters and relationships along the right-most branch (the collection subtree) of the scene graph, and **Visual Encodings** record encoded channels for rectangle marks. Note that in other types of visualizations, such correspondences remain similar. We next give a detailed explanation of the six semantic labels by walking through the example stacked bar chart (in Figure 1(c)).

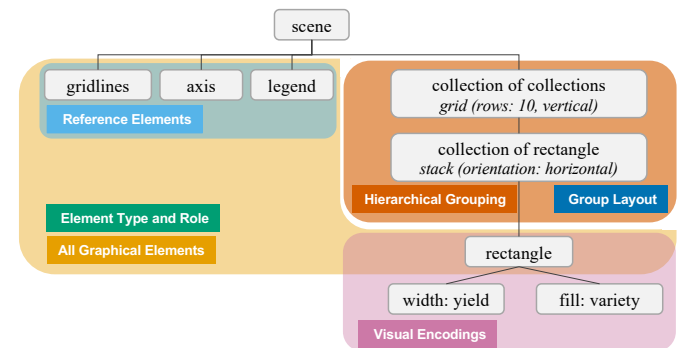


Fig. 4: The components in the stacked bar chart from Figure 1(c) and its correspondences to the labels in VISANATOMY.

All Graphic Primitives include every geometric shape and text element in an SVG chart (leaf nodes in the SVG hierarchy). Each element has several general properties, including element ID, element tag name, text content (if any), and fill color. Each element’s bounding box is expressed in absolute coordinates. Figure 1(c-I) presents three example primitives and their properties in the stacked bar chart.

Reference Elements are titles, axes, legends, and gridlines in a chart. Titles and gridlines are specified with the IDs of the corresponding SVG